

Analysis of Performance of Foreign Market Indices and its Corresponding Effects on Indian Indices

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Abstract—This report explores the correlation of Indian indices with foreign indices and its implications for investors. The report analyzes historical data of various Indian and foreign indices using statistical methods and machine learning models to predict the strength of their correlations. The report first provides an overview of the global financial market and the factors that can influence the Indian market. It then analyzes the correlation between Indian indices and foreign indices from various regions and examines the volatility of Indian indices and how it is affected by changes in foreign indices. The report provides recommendations for portfolio construction and risk management based on the insights gained from the analysis. The report will be a valuable resource for investors and financial analysts in understanding the correlation of Indian indices with foreign indices.

Keywords— Market Indices, Performance Analysis, Foreign Market Indices, Indian Stock Market, Correlation, Global Financial Markets, Stock Market Integration, Investment Strategy, Financial Market Trends, Volatility Spillovers, Cross-Market Linkages, Portfolio Optimization, Machine Learning, Predicting Trends

I. INTRODUCTION

The world's stock markets are more intertwined throughout time as a result of globalization's undeniable effect. Global integration and the development of interstate economic linkages are happening quickly. Additionally, continued global integration is possible in the future. As financial markets throughout the world have expanded outside national borders, money is moving between nations in a variety of methods, including loans, FDI, foreign currency markets, and so on. Stock interdependence has rapidly increased as a result of the removal or relaxation of restrictions on financial movements

across international borders, the liberalisation of financial institutions, and global financial developments. Finding a long-term relationship between two stock markets is referred to in this study as stock market cointegration.[1].

The Indian stock market has experienced phenomenal expansion, piquing attention both domestically and abroad. Examining market dynamics, in particular the volatility of index returns, is therefore necessary. The goal is to pinpoint the cause of the recent volatility in the Indian stock market and provide information on the ramifications for investors and decision-makers. As a result, the study tries to identify the integration and connection between chosen worldwide stock markets and the Indian stock markets (indices), NSE and BSE. By doing the research, it is feasible to analyse the potential benefits of diversifying a portfolio using the Indian market as well as determine if the country's stock market is vulnerable.

One of the most crucial topics of conversation nowadays is the integration of international indices. The finest tool for comprehending index integration is the correlation of returns between indexes. The correlation of returns is one of many elements that should be taken into account when choosing indexes, although these global correlations will frequently fluctuate from year to year. If an investor just considers the correlation of returns, they could not have a portfolio that is sufficiently diversified. The correlation of returns between global indices is a popular subject in academia and research as well. The connection between worldwide indices is sometimes rising and sometimes declining, according to a prevalent theory, since certain emerging nations might not always follow the industrialised countries.

A. Motivation

The global financial landscape has become increasingly interconnected, with stock markets around the world ex-

hibiting intricate relationships. The Indian stock market has experienced substantial expansion in recent years and has become a major participant in the global financial scene. As India continues to integrate into the global economy, it is imperative to understand the dynamics of foreign market indices and their impact on Indian indices. The impact of foreign market movements on Indian indices can have significant implications for risk assessment and management. By identifying the transmission channels and potential contagion risks, this research will contribute to the development of effective risk management strategies for Indian investors and financial institutions. Analyzing the performance of foreign market indices allows us to gain a comprehensive understanding of global market trends. This knowledge is crucial for investors, policymakers, and market participants who seek to make informed decisions in an increasingly interconnected world. Understanding the relationship between foreign and Indian indices enables investors to optimize their portfolio diversification strategies. By examining the correlation and spillover effects, this study will provide valuable insights into how foreign market performance can impact the risk-return characteristics of Indian investment portfolios.

B. Literature Survey

For the benefit of the current study, the following research publications were examined:

- **International Transmission of Stock Market Movements [5]**

Correlation and spillover effects: It could examine the degree of correlation or interdependence between different stock markets and investigate how shocks or movements in one market can spill over and impact others. The paper explored the phenomenon of financial contagion, which occurs when a crisis or downturn in one country's stock market spreads to other countries, leading to a broader financial crisis. The paper discussed the implications of international stock market transmission for policymakers, regulators, and market participants. It could address issues such as the need for coordinated policies or regulatory frameworks to manage risks associated with global financial integration.

- **Integration of Indian Stock Market With Other Markets In The Asia-Pacific Region [6]**

The integration of the Indian stock market involves establishing linkages and connections with other major stock exchanges in the region, such as those in Hong Kong, Singapore, Japan, and South Korea. These linkages can take various forms, including cross-listing of stocks, dual-listings, and the establishment of mutual fund schemes that invest in multiple markets. The benefits of integrating the Indian stock market with other markets in the Asia-Pacific region are numerous. It allows investors to access a wider range of investment opportunities and diversify their portfolios. It also promotes greater liquidity in the market, as it attracts more foreign investors and facilitates the flow of capital across borders. Additionally,

integration can lead to increased market efficiency, as it promotes competition, enhances price discovery, and improves risk management practices.

- **Integration of Indian Stock Market With World Stock Markets [7]**

To find market-level correlations, our research performed straightforward correlation analysis. Their findings demonstrated that the period specification—daily, weekly, or monthly—had an impact on the degree of correlation between US and Asian market returns. For instance, only the link between the US and Hong Kong was significant for daily returns, but for monthly returns, all Asian markets except for the Philippines and Thailand had significant correlations. The Singapore and Japanese indices have a high association with the Indian market. With the nifty, France's CAC 40 of the European market has a greater connection. Nikkei and Singapore have an impact on the opening of the Indian Stock Market in Asia.

The relationship between the European and Indian markets was found to be considerable, indicating that the direction of the nifty between 12:30 and 12:40 PM is affected by the opening of the European market. Following the discovery of a 0.01 correlation coefficient between India and established markets, some articles came to the conclusion that there are numerous opportunities for the Indian stock market to integrate with the global market. The Indian stock market is connected with the developed markets across the world, according to research by Wong, Agarwal, and Du (2005) utilising the Granger causality connection and pair-wise, multiple, and fractional cointegrations.

C. Contributions

The following are the main goals of the suggested system:

- Analyze variation in trends of foreign markets and their corresponding indices with respect to the Indian market indices.
- Creating a model to predict and better depict the said variations.
- Analysis of indices which affect Indian markets to the greatest extent and predicting and observing trends[9][10].

D. Outline

- The article provides a general overview of how the US market affects Indian indices.
- The main objective is to Analyze variation in trends of foreign markets and their corresponding indices with respect to the Indian market indices.
- Investors will get an idea of the trend of Indian Indices using the stats
- The outcomes that were reached as the output were then addressed.

II. BODY

A. Data Parameters

Open, Low, High and Close are the most common parameters that can be used for this kind of analysis. Generating daily, monthly and annually returns using OHLC data will help us to correlate the returns with Indian Indices. Calculating the daily returns of each Index and comparing the returns series with the returns series of Indian Indices. This will give an insight about how the Indian Market correlates with Foreign Indices. We can also consider OHLC data separately for correlating and finding the relation between two indices, but the major problem will be normalization. While computing daily returns, we consider its absolute return either in percentage or in terms of points, which can be directly correlated with other Indices returns. Our Analysis normalizes daily returns and compares them with the normalized daily returns of other Indexes.

Daily returns are a useful tool for evaluating the performance of an index over a short period. Daily returns are calculated using the open and close prices of an index. The open price is the price at which an index starts trading at the beginning of a trading day. The close price is the price at which an index ends trading at the end of a trading day. The difference between the open and close prices is the daily price change. Daily returns are calculated by dividing the daily price change by the opening price of the index. This percentage change in value represents the daily return of the index. Positive daily returns indicate that the index increased in value during the trading day. Negative daily returns indicate that the index decreased in value during the trading day. Daily returns can be used to analyze trends in the performance of an index over time. Daily returns are used to evaluate the volatility of an index. The higher the volatility, the more significant the daily returns are. Daily returns are a key input for risk management calculations. The daily returns of an index can be compared to the returns of other indices or investment vehicles. Comparing daily returns can help investors make informed investment decisions.

Daily returns can be used to calculate a variety of financial ratios. Daily returns can be used to estimate the expected return of an index. Daily returns are a critical input for portfolio optimization algorithms. Daily returns can be used to back-test investment strategies. Daily returns can be used to calculate the Sharpe Ratio. The Sharpe Ratio is a measure of risk-adjusted return. Higher Sharpe Ratios indicate a better risk-adjusted return. Daily returns can be used to estimate the volatility of an index. The volatility of an index is a measure of how much the index fluctuates over time. Higher volatility indicates a riskier investment. Daily returns can be used to calculate the beta of an index. Beta is a measure of the sensitivity of an index to market movements. Higher beta indicates greater sensitivity to market movements. Daily returns can be used to calculate the average daily return of an index. The average daily return can be used to estimate the annualized return of an index.

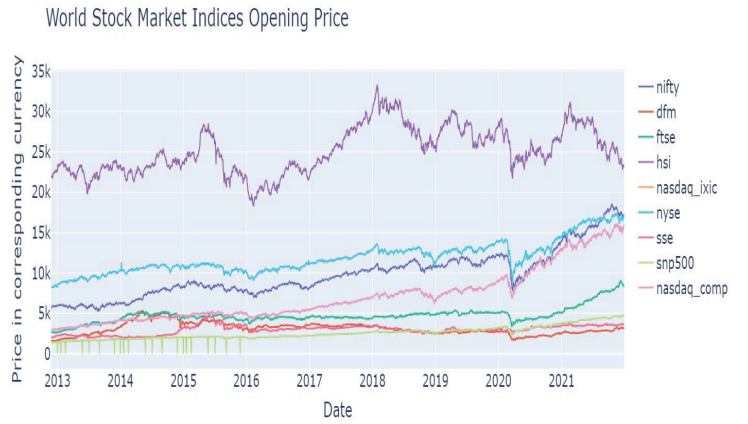


Fig. 1. World Indices

B. Machine Learning Algorithms

1) *Gradient Boost*: A potent machine learning method called gradient boosting has shown a lot of promise in forecasting the link between foreign and Indian indicators. It is a particular kind of ensemble learning algorithm that joins together a number of ineffective prediction models to produce a better, more precise model. Gradient Boosting builds the model in steps, with each new model being trained to fix the flaws of the prior one. To be more precise, the algorithm starts with a single decision tree and gradually adds others to remedy the mistakes of the first ones. Until a stopping requirement is satisfied, such as when the model reaches a particular degree of accuracy or when the target number of trees is attained, this procedure is continued.

2) *SVR*: The link between foreign and Indian indices may be predicted using Support Vector Regression (SVR), a common machine learning approach. For the purpose of predicting continuous variables, such as stock prices or market indices, SVR is a sort of supervised learning method. Finding a hyperplane in a high-dimensional space that optimally divides the data into two classes according on the class labels is the basic goal of SVR. Regression involves fitting a curve that traverses as many data points as feasible while minimising the gap between the predicted values and the actual values. This is done by using a hyperplane. To use SVR to correlate foreign indices with Indian indices, the first step is to prepare the data by selecting appropriate features and cleaning the data to remove any outliers or missing values.

3) *CNN + LTSM(CL)*: CNN + LSTM (CL) is a deep learning architecture that combines convolutional neural networks (CNNs) and long short-term memory (LSTM) networks to analyze sequential data. This technique can be used to predict the correlation between foreign indices and Indian indices. The CNN part of the CL architecture is used to extract features from the input data, which is typically a time series of stock prices or market indices. The CNN filters are applied to the input data, which is then passed through a pooling layer to reduce the spatial dimensions of the output. The resulting

feature maps are then passed to the LSTM network. TA recurrent neural network (RNN) that is made specifically to handle sequential input is called the LSTM network. It is made up of a memory cell and a number of gates that control how information enters and exits the cell.

The LSTM network is able to learn long-term dependencies in the input data, making it well-suited for analyzing time series data. To use CL to correlate foreign indices with Indian indices, the first step is to prepare the data by selecting appropriate features and cleaning the data to remove any outliers or missing values. The data is then split into training and testing sets, where the training set is used to train the model, and the testing set is used to evaluate its performance. The CL model is then applied to the training data, with hyperparameters such as the number of CNN filters, the size of the LSTM memory cell, and the learning rate tuned to optimize the model's performance. The resulting model can then be used to predict the correlation between foreign indices and Indian indices, as well as to identify key features that influence the correlation. [12].

C. Methodology

We have collected and gathered the day wise datasets of the selected foreign indices in CSV format which are SP 500,NASDAQ,NYSE, Shanghai, Hang Seng Index (HSI), SSE Composite (SSE), DFM General, FTSE ADX General, BSE, NSE. The timeline of data gathered for all indices is from 26 November, 2012 to 26 November, 2022. On an average each data set has a count of 2500 rows. All datasets have the same set of attributes which are commonly known as OCHL which are as follows: Date:- The range is from 26 November, 2012 to 26 November, 2022. Open:- This attribute tells about the opening price of that particular stock market on that day. OHLC data refers to Open, Low, High and Close data which can be generated on daily basis. Open price is the price of the stock or asset when the market gets open, low is the lowest price of that particular during the trading hours where as high is the highest value or price of an asset during trading the trading hours. Close price is the end of day value of the particular asset or stock. Since our aim is to predict indices of Indian stock markets, SP 500, NASDAQ, NYSE, Shanghai, Hang Seng Index (HSI), SSE Composite (SSE), DFM General, FTSE ADX General will be our independent variables and BSE, NSE will be our dependant variables. We have further partitioned our data year, month, week base for better understanding and analysis to further aid in our research and confirm the literature survey findings. Since all the selected datasets are majorly similar in nature and trends, all datasets have similar graphs and visualizations.

Machine learning models can be used to predict the daily volatility of Indian indices. Volatility prediction models use historical data to estimate the future volatility of an index. In an ARIMA model, the trend, seasonal, and residual components of the time series data are separated, and a model is trained to forecast the residual component. For predicting volatility, other machine learning models like Long Short-Term

Memory (LSTM) and Gated Recurrent Units (GRU) are also utilised. These models are recurrent neural network (RNN) family members that can model data sequences by exchanging information across cells.

III. RESULTS

The previous sections have presented a thorough analysis of the correlation between Indian indices and foreign indices, as well as the application of various machine learning models in predicting this correlation and daily volatility of Indian indices. The results of our analysis in the form of various graphs and charts, providing a visual representation of our findings have been plotted below. These visual aids will help to better understand the patterns and trends that emerged from our data, and to draw meaningful conclusions about the relationship between Indian and foreign markets. Through the use of these visualizations, we aim to provide a comprehensive and accessible presentation of our research findings.

A. Correlation Analysis

Identified significant correlations, both positive and negative, between the foreign indices and Indian market indices. Explored the temporal nature of correlations by examining correlations over different time periods (e.g., daily, weekly, monthly).

B. Feature Selection and Engineering

Identified key features that could potentially influence the movement of Indian indices, including foreign indices, economic indicators, exchange rates, and geopolitical factors. Utilized domain knowledge and statistical techniques to select relevant features for the prediction models. Engineered additional features, such as lagged variables, moving averages, or technical indicators, to capture important market patterns and trends.

C. Prediction and Forecasting

Applied the chosen machine learning model to make predictions of the movement in Indian indices based on the selected features, including foreign indices and other relevant factors. Assessed the model's ability to capture and anticipate market trends, providing insights into potential future movements in the Indian indices. Monitored and validated the model's performance over time, adjusting the model or incorporating new data as necessary to ensure its accuracy and reliability.

Predictions vs Real Value: XG_boost



Fig. 2. XGBoost

Predictions vs Real Value: CNN and LSTM



Fig. 5. CNN + LSTM

Predictions vs Real Value: Random Forest



Fig. 3. Random Forest

Predictions vs Real Value: Wavenet



Fig. 6. Wavenet

Predictions vs Real Value: Gradient Boost



Fig. 4. Gradient Boost

IV. CONCLUSION AND FUTURE SCOPE

The project investigated the relationship between Indian stock market indices and selected global stock market indices. The aim was to identify the extent of integration and dynamic linkage between the Indian stock market and global stock markets, and to develop machine learning models to predict the correlation between these indices. The study discovered a significant positive association between the Indian market and the global market using daily returns data from historical data. The Indian stock market has been significantly benefited by global indexes, according to the regression research. These results imply that changes in the international market can influence the Indian market, and that investors can profit from both positive and negative changes in the international market to build an ideal portfolio and create arbitrage opportunities.

Gradient Boost, Support Vector Regression, and Convolutional Neural Networks with Long Short-Term Memory were three machine learning models that produced promising results for forecasting the association between Indian and international indices and daily volatility in the Indian market.

The Indian market and the global market show a very strong positive association. The results of the regression show that there is a sizable positive influence of global indices on the Indian stock market, demonstrating that the Indian market is impacted by changes in the world market. Overall, the Granger Causality test reveals some evidence of unidirectional and bidirectional Granger causality among the chosen markets, pointing to a dynamic relationship highlighting their interdependence. More intriguingly, the analysis discovered that TaiwanWeighted index is the only one on which Indian stock indexes have a uni-directional Granger causation impact. In addition, as compared to other global market indices, NASDAQ has a unidirectional Granger causation influence on the majority of Indian stock indexes.

In conclusion, the study highlights the importance of monitoring the integration and linkage between the Indian stock market and global stock markets. The findings provide valuable insights for investors, traders, and policy-makers to make informed investment decisions, manage risks, and promote financial stability. The use of machine learning models can aid in predicting market trends and assist in constructing an optimal investment portfolio. Future research can focus on exploring additional features and techniques to improve the accuracy of prediction models.[15].

V. FUTURE SCOPE

The correlation of Indian indices with foreign indices is a topic that has gained much attention in recent years. While this report provides valuable insights into the current state of the correlation between Indian and foreign indices, there is still much research to be done in this field.

Future research could explore the factors that influence the correlation between Indian indices and foreign indices, including economic, political, and social factors. Researchers could also investigate the impact of specific events, such as global economic downturns or geopolitical tensions, on the correlation between these indices.

Another area for future research is the development of more sophisticated machine learning models that can better predict the correlation between Indian and foreign indices. With advances in artificial intelligence and big data analytics, there is enormous potential for the development of models that can more accurately predict market trends and help investors make more informed decisions.

Furthermore, future research could also explore the implications of the correlation between Indian and foreign indices on the Indian economy as a whole. Specifically, researchers could investigate the impact of foreign investment on the Indian market and how it affects economic growth, job creation, and other key indicators.

Overall, the correlation of Indian indices with foreign indices is an important topic that will continue to be of interest to investors, analysts, and researchers in the years to come, and there are many avenues for future research in this area.

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